Causal Symbolic regression and structure learning

IA pour la Science

April 6th, 2026

Causal structure learning with symbolic regression

The goal is to learn to search a model of optimal structure.

The structure must be admissible:

1. An acyclic directed graph with given number of roots and leafs.
2. Its nodes are operations from a given set.
3. It must fit the data and respect statistical assumptions about the data.

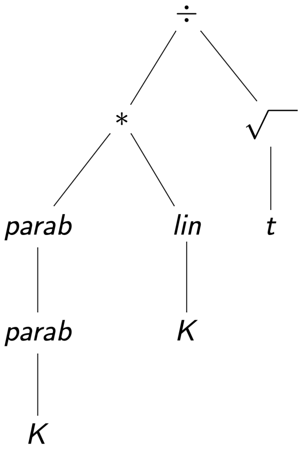
Two main approaches are:

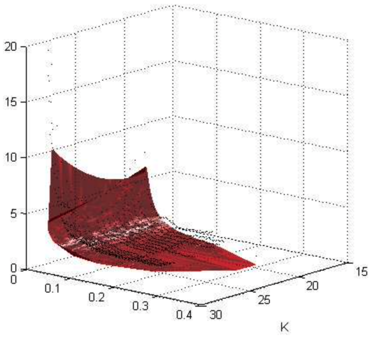
1. Symbolic regression is another computationally complex genetic algorithm.
2. Structure learning shows difficulties in delivering admissible structures of models.

The slides below lists variants of problem statement and methods to solve it.

Causal discovery for syblolic regression modeling

$$\sigma = \frac{(w_1 K + w_2)(w_1 K^2 + w_2 K + w_3)}{\sqrt{t}}$$





The set of basic functions $\mathfrak{G}$

There is given a set $\mathfrak{G} = \{\text{id}, g_1, \dots, g_l | g = g(\mathbf{b}, \mathbf{x}')\}$, that is, there are given

1) the function $g : (\mathbf{b}, \mathbf{x}') \mapsto \mathbf{x}''$,
2) its parameters $\mathbf{b}$,
3) arity $v(g)$ of the function g and an order of arguments,
4) a domain $\text{dom}(g)$ and a codomain $\text{cod}(g)$.

Consider a model $f(\mathbf{w}, \mathbf{x})$ given by a superposition

$$f(\mathbf{w}, \mathbf{x}) = (g_{i(1)} \circ \dots \circ g_{i(K)})(\mathbf{x}), \quad \mathbf{w} = [\mathbf{b}_{i(1)}^\mathsf{T}, \dots, \mathbf{b}_{i(K)}^\mathsf{T}]^\mathsf{T}.$$

An admissible superposition f

is a superposition such that

$$\text{cod}(g_{i(k+1)}) \subseteq \text{dom}(g_{i(k)}), \qquad k = 1, \dots, K-1.$$

Generation of the model set $\mathfrak{F}$

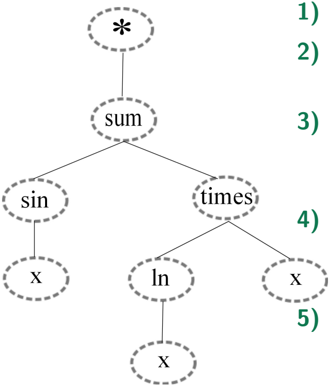
To generate the models we use

1) the set $\mathrm{dom}(\mathbf{x})$,
2) the set of basic functions $\mathfrak{G} = \{\mathrm{id}, g\}$, $g : \mathbf{x} \mapsto \mathbf{x}'$,
3) the set Gen of rules for superposition generation,
4) the set Rem of rules for isomorphic superpositions simplification and estimation.

We propose the following basic methods for the superpositions generation:

- inductive generation,
- structure learning,
- direct search.

Γ_f f:



$$f = \sin(x) + (\ln x)x$$

1) the root $*$ of the tree Γ_f has the single vertex,

2) other vertices V_i correspond to the functions $g_r \in \mathfrak{G}$: $V_i \mapsto g_r$,

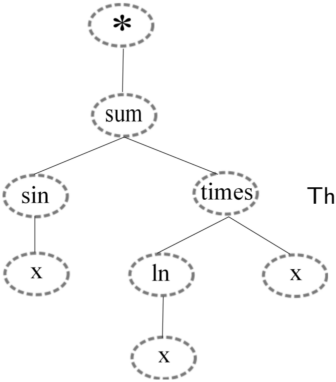
3) the number of children V_j of the vertex V_i equals to an arity of the corresponding function g_r: $\mathrm{val}(V_j) = v(g_{r(i)})$,

4) the domain of the function $g_{r(i)}$ of a child V_j contains the codomain of the function $g_{r(j)}$ of the parent V_i: $\mathrm{dom}(g_{r(i)}) \supseteq \mathrm{cod}(g_{r(j)})$,

5) an order of vertices traversal with a parent vertex V_i corresponds to the order of arguments of the corresponding function $g_{r(i)}$,

6) the leaves Γ_f correspond to the independent variables, elements of the vector $\mathbf{x}$.

Link matrix Z_f estimation limitations



$f = \sin(x) + (\ln x)x$

The link matrix $\mathbf{Z}_f$ for the tree Γ_f

	sum	times	ln	sin	x
$*$	1	0	0	0	0
sum	0	1	1	0	0
times	0	0	0	1	1
ln	0	0	0	0	1
sin	0	0	0	0	1

The link probability matrix $\mathbf{P}_f$ for the tree Γ_f

	sum	times	ln	sin	x
$*$	0.7	0.1	0.1	0.1	0.2
sum	0.2	0.7	0.8	0.1	0.2
times	0.1	0.3	0	0.8	0.8
ln	0.2	0.1	0.3	0.1	0.9
sin	0.1	0.2	0.1	0	0.8

$\mathfrak{Z}$ is a set of matrices corresponding to the superpositions from $\mathfrak{F}$.

Structure learning problem

There is given a sample $\mathfrak{D} = \{(\mathbf{D}_k, f_k)\}$ where the element $\mathbf{D}_k = (\underset{m\times n}{\mathbf{X}}, \underset{m\times 1}{\mathbf{y}})$, there given $\mathfrak{G}$ and $\mathfrak{F} = \{f_s \mid \mathbf{f}_s : (\hat{\mathbf{w}}_k, \mathbf{X}) \mapsto \mathbf{y}, s \in \mathbb{N}\}$.

The goal

to find an algorithm $a : \mathbf{D}_k \mapsto f_s$ following the condition

$$\mathbf{Z}_{f_s} = \arg\max_{\mathbf{Z}\in\mathfrak{Z}} \sum_{i,j} P_{ij} \times Z_{i,j}.$$

The index $\hat{s}$, $f_{\hat{s}}$ provides a minimum for the error function S:

$$\hat{s} = \arg\min_{s\in\{1,\dots,|\mathfrak{F}|\}} S(f_s \mid \hat{\mathbf{w}}_k, \mathbf{D}_k),$$

where $\hat{\mathbf{w}}_k$ is an optimal vector of parameters f_s for each $f_s \in \mathfrak{F}$ with the fixed $\mathbf{D}_k$:

$$\hat{\mathbf{w}}_k = \arg\min_{\mathbf{w}\in\mathbb{W}_s} S(\mathbf{w} \mid f_s, \mathbf{D}_k).$$

Problem statement for machine learning

Formal problem statement, an analyst has to set

1) an algebraic structure for the dataset from measurements
2) a data generation hypothesis from 1)
3) a model, or a mixture from 2)
4) an error function (quality criteria with restrictions) from 2)
5) an optimization algorithm from 3) and 4)

The result of the model construction is a Cartesian product

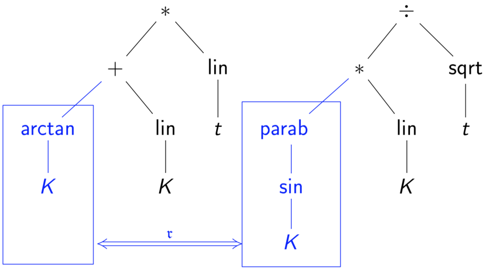
$$\{\textbf{models} \times \textbf{data sets} \times \textbf{quality critea}\}.$$

Def: *Big data rejects the i.i.d. (independent and identically distributed random variables) data generation hypothesis from 2). It requests a mixture model.*

Genetic optimization constructs symbolic regression model structure

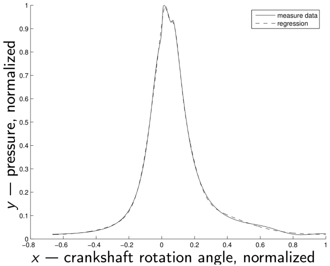
To create a model as a superposition of primitive functions

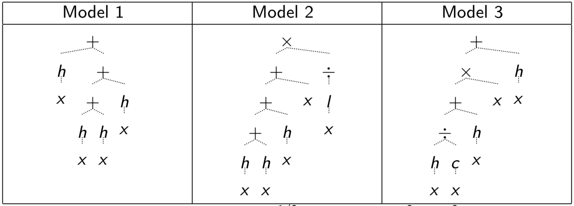
1) exchange random sub-trees between two models,
2) replace a random primitive for another one,
3) select the best models and repeat.



Simple superposition has 14 parameters wersus 2-NN has 64 parameters

Approximate the pressure in the combusting camera of a diesel engine





Legend: h — gaussian $y = \lambda(2\pi\sigma^{-1/2})\exp(-(x-\xi)^2(2\sigma^{-2})+a)$,
c — cubic $y = ax^3 + bx^2 + cx + d$, l — linear $y = ax + b$.

$$f_2 = g_1(g_2(g_3(g_4(g_5(x), g_6(x)), g_7(x)), x), g_8(x)).$$

The full representation of the Model 2 is

$$y = (ax+b)^{-1}\left(x + \sum_{i=1}^{3} \frac{\lambda_i}{\sqrt{2\pi}\sigma_i}\exp\left(-\frac{(x-\xi_i)^2}{2\sigma_i^2}\right) + a_i\right).$$

Time series samples for physical activity monitoring

Instances

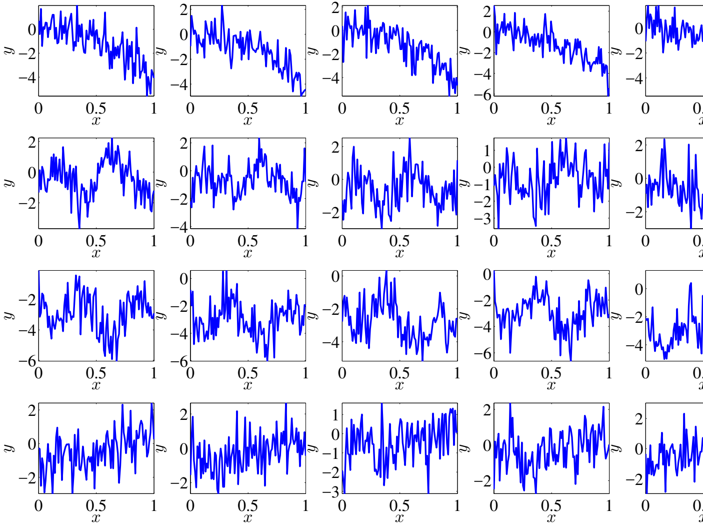
Classes

One

Two

Three

Four



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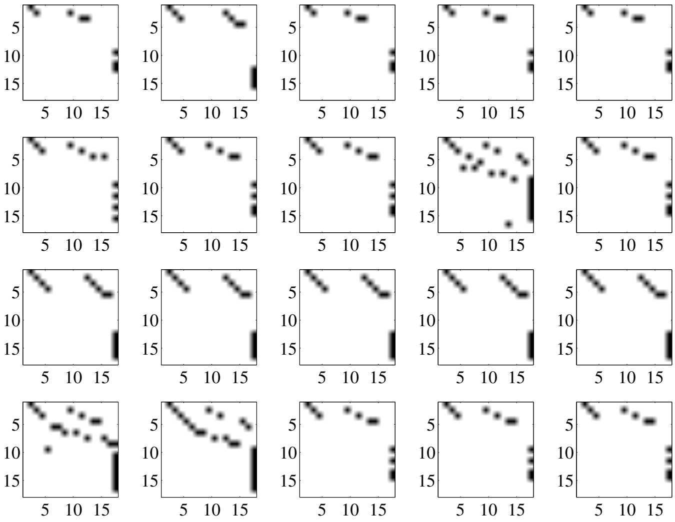
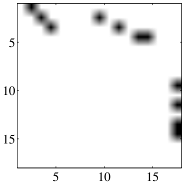
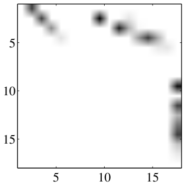


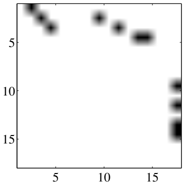
The initial and the forecasted superposition



Ground truth

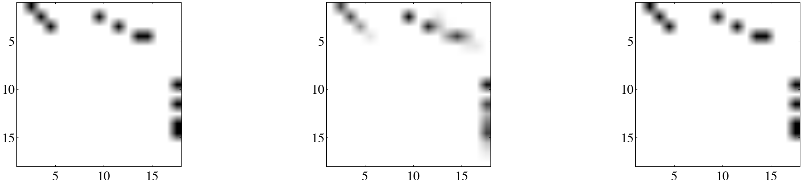


Forecasted probabilities



Forecasted superposition tree (model)

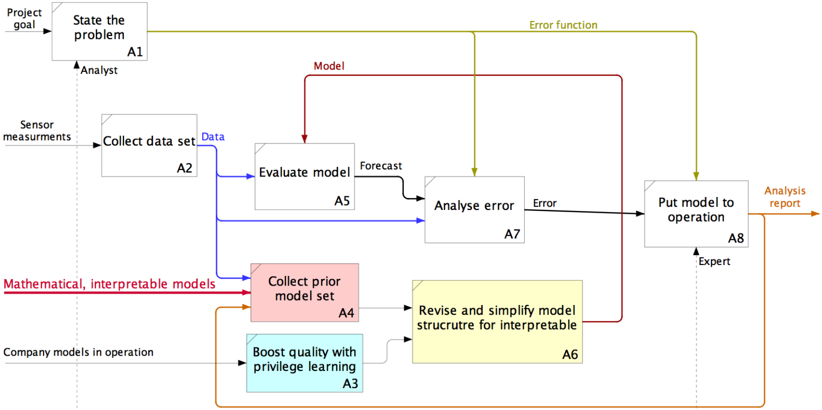
Initial and forecasted superposition



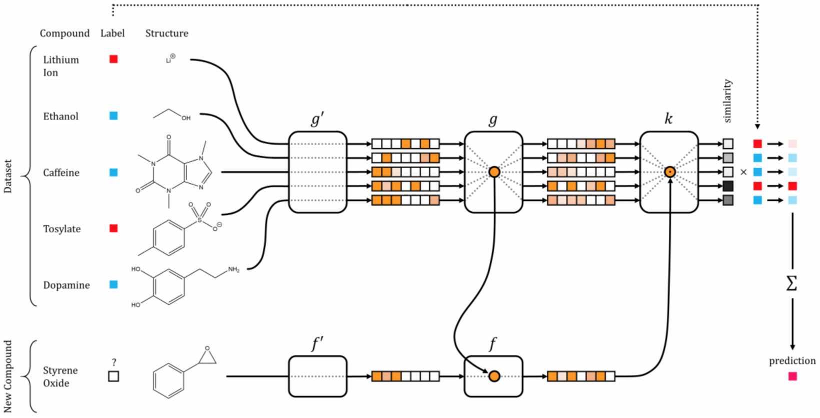
$$f = w_1 \cos(w_2 x + w_3) + w_4 x + w_5 \ln(w_6 x + w_7) + w_8,$$

$$f = \cos(x) + x + \ln(x), \qquad \mathbf{w} = [1, 1, 0, 1, 1, 1, 0, 0]^{\mathsf{T}}.$$

Put interpretable models to operation along with privilege learning models



Altae-Tran et al. 2016. Low data drug discovery with one-shot learning // arXiv



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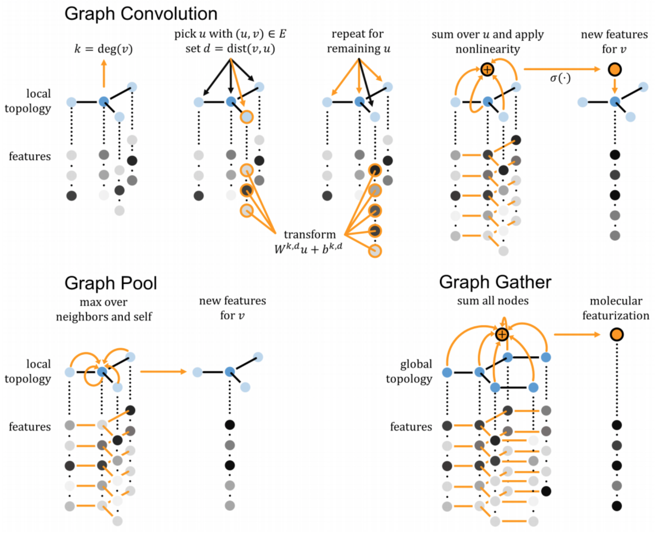
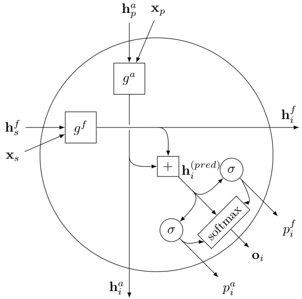
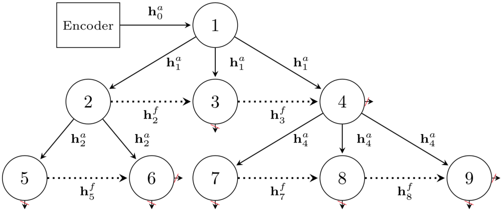


Alvarez-Melis, Jaakkola. 2017. Tree-structured decoding with doubly-recurrent neural networks // ICLR



A cell of the doubly-recurrent neural network corresponding to node *i* with parent *p* and sibling *s*.



Structure-unrolled DRNN network in an encoder-decoder setting. The nodes are labeled in the order in which they are generated. Solid (dashed) lines indicate ancestral (fraternal) connections. Crossed arrows indicate production halted by the topology modules.

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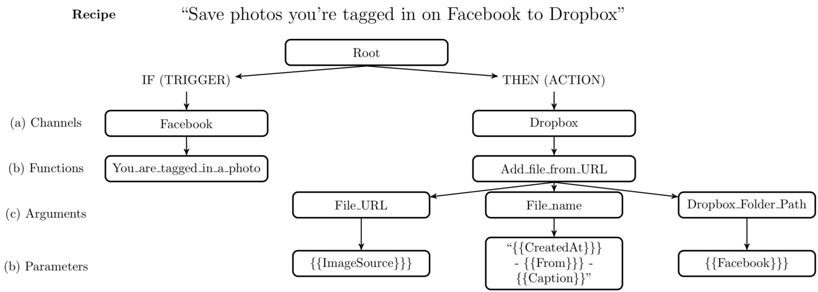

Example recipe from the IFTTT dataset. The description (above) is a user-generated natural language explanation of the if-this-then-that program (below).